

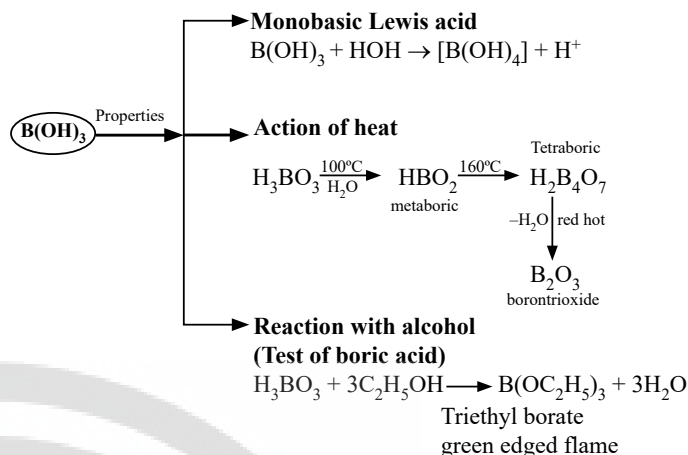
p-block Elements (Group 13 and 14)

SHORT NOTES

Properties of Group 13 Elements

Property	Remarks
Atomic Radii	$B < Al > Ga < In < Tl$
Ionisation Energy	$B > Al < Ga > In < Tl$
Electronegativity	$B > Al < Ga < In < Tl$
Oxidation state	General oxidation states: +1, +3 Stability of +1 state increases down the group due to inert pair effect
Metallic character	Increases down the group
Melting point	Decreases up to Ga then increases
Boiling point	Decreases down the group
Density	Increases down the group
Reactivity with air	Reacts with air to form oxide (at moderate temperature) and nitride (at high temperature) Crystalline boron unreactive, amorphous boron reacts. Aluminium forms oxide layer and does not react further.
Reactivity with acids and alkalis	Boron does not react. Aluminium reacts with both acids and alkalis to liberate hydrogen gas. Aluminium passive with conc. nitric acid.
Reactivity with halogens	Reacts with halogens to form trihalides (except triiodide of Thallium)
Nature of trioxides	Boron-acidic, Al and Ga - amphoteric, In and Tl-Basic

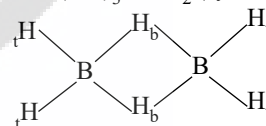
Orthoboric Acid [H_3BO_3 or $B(OH)_3$]



Diborane, B_2H_6

Properties

- Diborane is a colourless, highly toxic gas with a boiling point of 183 K.
- Diborane catches fire spontaneously upon exposure to air.
- Reaction with oxygen:
 $B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$; $\Delta_c H^\circ = -1976 \text{ kJ mol}^{-1}$
- Hydrolysis:
 $B_2H_6(g) + 6H_2O(l) \text{ (Cold is enough)} \rightarrow 2B(OH)_3(aq) + 6H_2(g)$
 $B_2H_6 + HCl \text{ (dry)} \xrightarrow[AlCl_3]{anh.} B_2H_5Cl + H_2$
- Reaction with Lewis Bases
 $B_2H_3 + 2NMe_3 \rightarrow 2BH_3 \cdot NMe_3$
 $B_2H_6 + 2CO \rightarrow 2BH_3 \cdot CO$
- $3B_2H_6 + 6NH_3 \rightarrow 3[BH_2(NH_3)_2]^+[BH_4]^- \xrightarrow{Heat}$
 unsymmetrical cleavage $2B_3N_3H_6 + 12H_2$
- Reaction with ROH:
 $B_2H_6 + 6ROH \rightarrow 2B(OR)_3 + 6H_2$ (symmetrical cleavage)

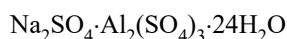


Dimer due to formation of 3 centre-2e⁻ bond inorganic benzene.

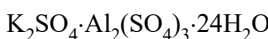
Alums [$M_2SO_4 \cdot M_2'(SO_4)_3 \cdot 24H_2O$]

Alums are the double salt of Type $[M_2SO_4 \cdot M_2'(SO_4)_3 \cdot 24H_2O]$
M : Monovalent Cation: $Na^+, K^+, Rb^+, Cs^+, NH_4^+$
M' : Trivalent Cation: $Al^{3+}, Fe^{3+} \text{ \& } Cr^{3+}$

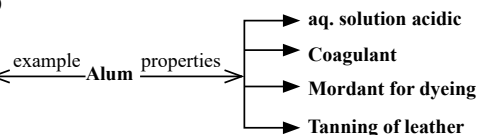
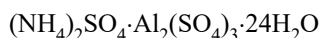
Soda alum



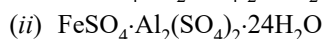
Potash alum



Ammonium alum



Pseudo alums are double sulphates of a divalent and a trivalent metals which crystallize with twenty four water molecules of crystallization. For example:



❖ They are not isomorphous with true alums.

Properties of Group 14 Elements

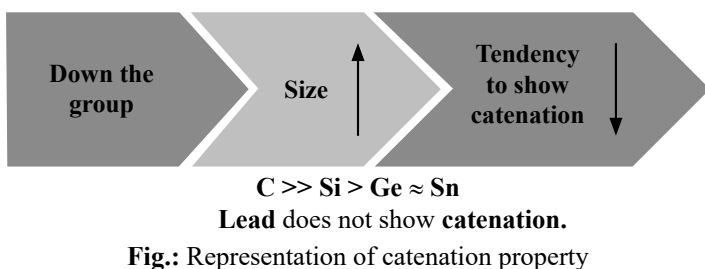
Property	Remarks
Covalent radii	$\text{C} < \text{Si} < \text{Ge} < \text{Sn} < \text{Pb}$
Ionisation Enthalpy	$\text{C} > \text{Si} > \text{Ge} < \text{Sn} > \text{Pb}$
Electronegativity	$\text{C} > \text{Pb} > \text{Si} \approx \text{Ge} \approx \text{Sn}$
Metallic character	Increases down the Group
Melting point Oxidation state	General oxidation state: +4 and +2 Stability of +2 oxidation state increases down the group due to inert pair effect
Reactivity towards oxygen	React with oxygen to form oxides (mainly two types of oxides, monoxide and dioxide)
Reactivity towards Water	C, Si, Ge and Pb do not react with water. Sn decomposes steam to form dioxide and dihydrogen gas. $\text{Sn} + 2\text{H}_2\text{O} \rightarrow \text{SnO}_2 + 2\text{H}_2$
Reactivity of halogen	These element can form halides of formula MX_2 and MX_4 (where X = F, Cl, Br, I) Hence members Ge to Pb are able to make halides of formula MX_2

Anomalous Behavior of Carbon

Carbon atoms have the tendency to link with one another through covalent bonds to form chains and rings. This property is called catenation. This is because C–C bonds are very strong. Down the group, as size increases tendency to show catenation decreases.

Table: Catenation Property

Bond	Bond enthalpy (kJ/mol^{-1})
C-C	348
Si-Si	297
Ge-Ge	260
Sn-Sn	240



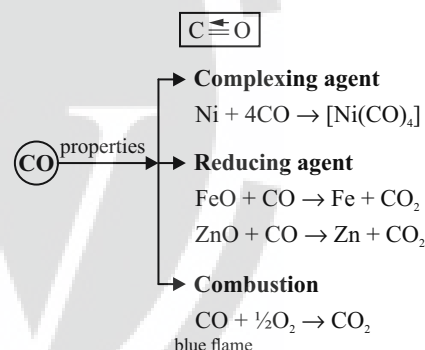
Allotropes of Carbon

Diamond, Graphite and Fullerenes.

- ❖ In diamond, each carbon atom undergoes sp^3 hybridisation and is linked to four other carbon atoms by using hybridized orbitals in tetrahedral manner. It is used as an abrasive for sharpening hard tools in making dies (die casting) and in the manufacture of tungsten filament for electric light bulbs.
- ❖ In Graphite, each carbon atom in hexagonal ring undergoes sp^2 hybridisation and make three sigma bonds with three neighboring carbon atoms. Fourth electron forms a π bond. The electrons are delocalised over the whole sheet. Electrons are mobile and, therefore graphite conducts electricity along the sheet.
- ❖ Fullerenes are made by the heating of graphite in an electrical arc in the presence of inert gases such as helium or argon. All the carbon atoms are equal and they undergo sp^2 hybridisation.

Carbon Monoxide [CO]

Colorless odourless, tasteless, neutral, poisonous gas.



Silicones (Organo Silicone Polymer)

- ❖ Silicones are polymeric organosilicon compounds containing Si–O–Si linkage. They have high thermal stability of Si–O–Si chains and are also called high temperature polymers.

General formula: $(\text{R}_2\text{SiO})_n$. Where R = $-\text{CH}_3$, $-\text{C}_2\text{H}_5$, $-\text{C}_6\text{H}_5$

